A RELATIVE DUALIZING COMPLEX IS UNIQUE UP TO ISOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The evaluation-tensor isomorphism for internal Homs. Invoked inside the counted proof (source lines 5645-5649) as a cross-chapter pointer. Rewrites the tensor of one dualizing complex into the internal Hom expression.

Source reference. Original reference more-algebra-lemma-internal-hom-evaluate-tensor-isomorphism: sources/stacks-dualizing/pinned-original-source.tex, source lines 2741–2741. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-internal-hom-evaluate-tensor-isomorphism, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $R \to A$ be a flat ring map of finite presentation. Any two relative dualizing complexes for $R \to A$ are isomorphic.*

Proof. Let K and L be two relative dualizing complexes for $R \to A$. Denote $K_1 = K \otimes^{\mathbf{L}}_A (A \otimes_R A)$ and $L_2 = (A \otimes_R A) \otimes^{\mathbf{L}}_A L$ the derived base changes via the first and second coprojections $A \to A \otimes_R A$. By symmetry the assumption on L_2 implies that $R\mathrm{Hom}_{A\otimes_R A}(A, L_2)$ is isomorphic to A. By More on Algebra, Lemma more-algebra-lemma-internal-hom-evaluate-tensor-isomorphism part (3) applied twice we have

$$A \otimes^{\mathbf{L}}_{A\otimes_R A} L_2 \cong R\mathrm{Hom}_{A\otimes_R A}(A, K_1 \otimes^{\mathbf{L}}_{A\otimes_R A} L_2) \cong A \otimes^{\mathbf{L}}_{A\otimes_R A} K_1$$

Applying the restriction functor $D(A \otimes_R A) \to D(A)$ for either coprojection we obtain the desired result. □

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.